

Pro-Tip: If you use Meraki, captive portals can integrate with Facebook Wi-Fi. But toggle “Collect email” off unless you want the compliance headache of storing PII.

4) VPN bridging between HQ & Home: IPSec / OpenVPN walkthrough

Work-from-anywhere reality

Post-pandemic, 67 % of Indian tech employees work remotely at least twice a week. Every unencrypted coffee-shop session is a sniffing opportunity, so pipe all traffic through a corporate VPN.



Choosing a protocol

- **IPSec IKEv2:** Fast reconnect, native in Windows/macOS/iOS; NordLayer offers automated site-to-site panels.
- **OpenVPN:** Runs on anything, easier to NAT-traverse, but ~10% slower.
- **WireGuard:** Blazing, but fewer GUI tools.

Scenario A: Head-Office ↔ Branch (Site-to-Site IPSec with NordLayer)

1. **Order a Dedicated Gateway** in NordLayer console; note [GW-IP](#).
2. **HO Router (e.g., FortiGate):** VPN ► IPSec Tunnels ► Create New ► Custom.
 - **Remote GW:** GW-IP
 - **Preshared Key:** generate 24-char random.
 - **Local Subnets:** 192.168.10.0/24; Remote Subnets: 10.6.0.0/20 (Nord default).
 - **Phase 1:** AES-256-SHA256, DH-Group 14, Lifespan 28 800 s.
3. **Branch Router (UniFi USG):** Networks ► Create New ► Site-to-Site VPN ► Manual IPsec.
 - Mirror settings, insert same preshared key.
4. **Firewall Rule:** Allow [Corp-LAN](#) ↔ [Branch-LAN](#) TCP/443, UDP/500, UDP/4500.
5. **Verify:** `diag vpn ike gateway list` (Forti); status should be “created”.

Scenario B: Individual remote worker (OpenVPN + pfSense)

1. **pfSense:** VPN ► OpenVPN Wizard; pick local CA, add server cert.
2. **Tunnel Network:** [10.8.0.0/24](#); Redirect Gateway: enabled to push all Internet traffic.
3. **Auth:** Username + Password + TLS-Auth key.
4. **Export Client Bundle:** send [.ovpn](#) to employee; they import into OpenVPN Connect (mobile/desktop).
5. **Split DNS:** In Advanced Push Options add [dhcp-option DOMAIN corp.local](#).
6. **Test:** From home PC, [tracert 8.8.8.8](#) should show one hop – the pfSense WAN IP.

Performance tuning

- **AES-GCM on CPUs with AES-NI** yields 650 Mb/s on mid-range x86 firewall.
- **MTU Clamp to 1350** if users report packet-loss on LTE.

Security must-do's

1. **MFA everywhere:** NordLayer dropped IKEv2 inside its apps after 1 June 2024; manual IKEv2 remains but enforce TOTP for logins.
2. **Rotate PSK quarterly;** add script to remind via cron.
3. **Log & Alert:** Push VPN events to your SIEM; CERT-In's 2024 advisory mandates 180-day log retention for VPNs.

5) Policy templates for BYOD, IoT Sensors & temporary contractors

BYOD – Bring Your Own Device

A 2024 Nasscom survey shows 74 % of Indian SMEs allow employees to use personal phones for email. Without rules, that's malware waiting to ride home. Adopt these clauses (adapted from free templates):

Clause	Must-Have Text
Device Eligibility	"Android 11 + with monthly security patches; iOS 15 +."
Security Controls	"Company MDM installs work profile, enforces PIN, remote-wipe consent."
Data Split	"Corporate data stored in AES-256 container; copy-paste to personal apps blocked."
Cost Sharing	"Company reimburses ₹250/month data on presentation of bill."
Exit Process	"Off-boarding triggers automatic wipe of the corporate container within 2 hours."